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Maths puzzles for early finishers

# Early Finisher Puzzles

## Primary 6

## Semester 1

Terms 1 and 2

20 single-page puzzles
Almost every puzzle checks itself
One star and two star pages
Teacher answer key included

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# Teacher notes (DRAFT)

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## What this is

Twenty single-page maths puzzles for Primary 6 pupils who finish their work early. This is Semester 1 of two booklets (Terms 1 and 2). Each page stands alone, so you can hand out one page or keep a tray of pages. The pupil needs no help to start.

## Which terms

The pages are grouped by the order in which Primary 6 topics are typically taught: this booklet follows Terms 1 and 2, the other booklet follows Semester 2 (Terms 3 and 4). Schools vary, so check your scheme of work: use a page whenever its topic has been taught, from either booklet.

## Stars

One star pages practise a topic as it is usually taught (10 pages). Two star pages use the same topics with two steps or two clues to combine (10 pages).

## How the pages check themselves

The boxed strip at the foot of each page tells the pupil how to check. The check shows that something is wrong. It does not give the answer, and it asks the pupil to see whether the answer makes sense. Every answer box and every letter box must be filled: a box left blank gives no letter, and the check cannot see it.

- **Code word.** The answers give letters that spell a word. The strip gives no hint of the word. Each word can be spelt from its letters in one way only, so answers in the wrong boxes never make another word, and one or two wrong answers very rarely do. A box left empty gives no letter and is not seen by the check.
- **Colour by code.** Each answer is found in a key of five numbers and the part is coloured with that crayon. The pupil counts the parts of each colour and compares with the box on the page; matching parts must share a colour.
- **Route.** The line must reach FINISH through the stated number of boxes, and every box on it must follow the rule on the page.
- **Letters in an order, a grid or a choice.** Each number, card, picture or choice carries a letter; the letters, read in the order asked, must spell a word. Where one item is left out, its letter is a decoy that is not part of the word.
- **Loop.** The numbers on touching tiles must match all the way round the loop.

Marking is not needed. To spot-check, look only at the check strip.

## How the checks were tested

A script tried every single wrong answer and every wrong arrangement on every page. Each check rejected at least 95% of them. A wrong answer that spells any of about 207 000 English words of two to nine letters counts as not rejected. Two wrong answers at once were not all tested.

# Teacher notes: using the pages (DRAFT)

## Using the pages

- **Time.** Most pages take 5 to 15 minutes; the colour-by-code pages (S1-09) take longer because pupils colour a picture. Pupils need their own crayons (five colours: yellow, red, blue, green, orange). For a pencil-only copy the pupil writes Y, R, B, G or O in each part; the tally and the matching parts are checked the same way. Light crayons work best: a dark crayon hides the number in the part on a photocopy.
- **If the check fails.** The pupil goes back, looks for the slip, and asks a friend or you if still stuck.
- **Stars.** One star: one skill, as usually taught. Two stars: the same topics with two steps or two clues to combine.
- **Drawings.** Pictures of circles, solids, graphs and angle figures are original drawings, drawn accurately to help the pupil see the shape. Pupils should work out the answer, not measure it. Pictures are not actual size.
- **Notation.** Ratios are written 3 : 5 and answers in simplest form. Where a page uses π it says whether to take 3.14 or 22/7 (22 over 7). Letters such as n, p and a stand for numbers; there are no brackets with letters. Money is written $12.50. Litres are written ℓ and cube centimetres cm³.
- **Rough paper and calculators.** Give the pupil rough paper for working. Every number is chosen so that the working can be done by hand; a pupil may use a calculator if your school allows it, but the check does not need one.
- **Revision pages.** Pages whose key tag begins with Revisits practise outcomes from Primary 4 or 5 that Primary 6 pupils meet again in revision. Primary 6 content only is practised on every other page.

## Photocopying

All pages are black and white with solid lines, so they copy cleanly. Use normal toner, not a light setting. Each page is one side only. Pupils write on the page: the boxes are large. Use A4 and print at 100% (actual size).

Pictures are not actual size.

# Contents: Semester 1


| Page | Code | Puzzle | Tier |
|---|---|---|---|
| S1-01 | P6-01 | Fraction Shares | 1 star |
| S1-02 | P6-02 | Dividing by Fractions | 1 star |
| S1-03 | P6-03 | Fraction Stories | 2 stars |
| S1-04 | P6-04 | Equal Ratios | 1 star |
| S1-05 | P6-05 | Missing Terms | 1 star |
| S1-06 | P6-06 | Share in a Ratio | 1 star |
| S1-07 | P6-07 | Three Quantities | 2 stars |
| S1-08 | P6-08 | Fractions and Ratios | 2 stars |
| S1-09 | P6-09 | Colour the Car | 2 stars |
| S1-10 | P6-10 | Find the Whole | 1 star |
| S1-11 | P6-11 | Percent Up and Down | 1 star |
| S1-12 | P6-12 | Percent Stories | 2 stars |
| S1-13 | P6-13 | Letters for Numbers | 1 star |
| S1-14 | P6-14 | Simplify | 1 star |
| S1-15 | P6-15 | Substitute | 2 stars |
| S1-16 | P6-16 | Find the Letter | 2 stars |
| S1-17 | P6-17 | Round and Round | 1 star |
| S1-18 | P6-18 | Circle Areas | 2 stars |
| S1-19 | P6-19 | Half and Quarter | 2 stars |
| S1-20 | P6-20 | Curves and Corners | 2 stars |


Pages are independent. The footer code shows the page in this booklet and, in brackets, its number in the full set of 40 (P6-01 to P6-40).

## What the booklet covers

**Fractions 1.1** dividing a proper fraction by a whole number **S1-01, S1-03**

**Fractions 1.2** dividing a whole number or a proper fraction by a proper fraction **S1-02, S1-03**

**Percentage 1.1** finding the whole given a part and the percentage **S1-09, S1-10, S1-12**

**Percentage 1.2** percentage increase or decrease **S1-09, S1-11, S1-12**

**Ratio 1.1** ratio a:b and a:b:c with whole numbers **S1-04, S1-07**

**Ratio 1.2** equivalent ratios **S1-04, S1-05, S1-09**

**Ratio 1.3** dividing a quantity in a given ratio **S1-06, S1-09**

**Ratio 1.4** ratio in simplest form **S1-04, S1-07**

**Ratio 1.5** ratio of two or three quantities **S1-06, S1-07, S1-08**

**Ratio 1.6** finding the missing term in a pair of equivalent ratios **S1-05, S1-09**

**Ratio 1.7** relationship between fraction and ratio **S1-08**

**Algebra 1.1** using a letter to stand for an unknown number **S1-13**

**Algebra 1.2** notation of simple expressions such as a + 3, a - 3, 3a and a ÷ 3 **S1-13, S1-15**

**Algebra 1.3** simplifying simple linear expressions (no brackets) **S1-14, S1-16**

**Algebra 1.4** evaluating a simple expression by substitution **S1-15**

**Algebra 1.5** solving simple linear equations with whole-number coefficients **S1-16**

**Area and Volume 1.1** area and circumference of a circle **S1-17, S1-18**

**Area and Volume 1.2** area and perimeter of a semicircle and a quarter circle **S1-19, S1-20**

**Area and Volume 1.3** composite figures made of squares, rectangles, triangles, semicircles and quarter circles **S1-20**

Outcomes practised in the other booklet: Area and Volume 2.1; Area and Volume 2.2; Area and Volume 2.3; Area and Volume 2.4; Area and Volume 2.5; Geometry 1.1; Statistics 1.1; Statistics 1.2; Revisits Primary 4 Statistics; Revisits Primary 4 Geometry; Revisits Primary 4 Decimals; Revisits Primary 5 Fractions; Revisits Primary 5 Percentage; Revisits Primary 5 Rate; Revisits Primary 5 Volume.

S1-01 (P6-01)

# Fraction Shares

★ Name _____

**Topic: Dividing a fraction by a whole number**

Divide. Write each answer in simplest form.
Use the code table.

**1** Grace pours $\frac{2}{5}\ \ell$ of juice equally into 5 cups. How much juice is in each cup?

$\square\ \ell$

**2** $\frac{9}{13} \div 3 = \frac{\square}{\square}$

**3** $\frac{1}{3} \div 9 = \frac{\square}{\square}$

**4** $\frac{4}{5} \div 6 = \frac{\square}{\square}$

**5** A rope $\frac{5}{9}$ m long is cut into 3 equal pieces. How long is each piece?

$\square$ m

**6** $\frac{3}{8} \div 7 = \frac{\square}{\square}$

**Code table**

| | | | | | |
|---|---|---|---|---|---|
| $\frac{1}{27}$ = E | $\frac{2}{15}$ = A | $\frac{2}{25}$ = C | $\frac{3}{13}$ = R | $\frac{3}{56}$ = Y | $\frac{4}{56}$ = P |
| $\frac{5}{27}$ = M | $\frac{5}{28}$ = Q | $\frac{9}{40}$ = V | | | |

**Check it.** Write the letters in order. Do they spell a word?

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S1-02 (P6-02)

# Dividing by Fractions

★ Name _____

**Topic: Dividing a whole number or a fraction by a fraction**

Start at START. Move up, down, left or right.
Step only on boxes where the answer is 6.
Do not step on a box twice.

| | | | |
|---|---|---|---|
| $10 \div \frac{3}{5}$ | START $2 \div \frac{1}{3}$ | $\frac{6}{11} \div \frac{1}{11}$ | $\frac{6}{7} \div \frac{1}{7}$ |
| $2 \div \frac{2}{7}$ | $3 \div \frac{9}{10}$ | $\frac{1}{2} \div \frac{1}{12}$ | $\frac{8}{9} \div \frac{1}{7}$ |
| FINISH $\frac{3}{4} \div \frac{1}{8}$ | $\frac{7}{9} \div \frac{2}{11}$ | $1 \div \frac{1}{6}$ | $3 \div \frac{5}{12}$ |
| $5 \div \frac{5}{6}$ | $4 \div \frac{2}{3}$ | $\frac{2}{3} \div \frac{1}{9}$ | $3 \div \frac{1}{2}$ |

**Check it.** Does your line reach FINISH? Count its boxes: are there 8? Does every box equal 6?

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S1-03 (P6-03)

# Fraction Stories

★★ Name _____

**Topic: Fraction stories: dividing by a whole number or by a fraction**

Work out each answer.
Circle the letter beside the right answer.
Write the six letters in the boxes.

**1** A jug has $\frac{1}{3}\ \ell$ of milk. Each glass takes $\frac{1}{6}\ \ell$. How many glasses can be filled?

(M) 2 (S) $\frac{1}{18}$ (Z) $\frac{1}{2}$

**2** $\frac{8}{9}$ of the garden is split into 4 equal portions for 4 children. What fraction of the whole garden is one portion?

(O) $\frac{2}{9}$ (Q) $1\frac{5}{36}$ (T) $\frac{1}{4}$

**3** Kumar has 6 m of ribbon. Each piece must be $\frac{2}{3}$ m long. How many pieces can Kumar cut?

(F) $\frac{1}{9}$ (G) 4 (N) 9

**4** $\frac{1}{5}$ of the cake is split into 2 equal portions for 2 children. What fraction of the whole cake is one portion?

(V) $\frac{1}{2}$ (A) $\frac{2}{5}$ (K) $\frac{1}{10}$

**5** Hui has 2 m of ribbon. Each piece must be $\frac{2}{3}$ m long. How many pieces can Hui cut?

(E) 3 (X) $1\frac{1}{3}$ (W) $\frac{1}{3}$

**6** A jug has $\frac{5}{8}\ \ell$ of milk. Each glass takes $\frac{1}{8}\ \ell$. How many glasses can be filled?

(Y) 5 (I) $\frac{1}{5}$ (C) $\frac{5}{64}$

**Check it.** Write the six letters in order. Do they spell a word?

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S1-04 (P6-04)

# Equal Ratios

Name _____

**Topic: Ratios: equivalent ratios and simplest form**

Match each ratio to an equal ratio.
Write the card letter in the box.
Use each card once. Two cards are not used.

**Items**

| | |
|---|---|
| 1 **12 : 21 : 18** ☐ | 2 **10 : 6** ☐ |
| 3 **14 : 18** ☐ | 4 **30 : 48** ☐ |
| 5 **27 : 24 : 6** ☐ | 6 **20 : 45** ☐ |

**Cards**

| | | | |
|---|---|---|---|
| X **18 : 7 : 8** | M **18 : 16 : 4** | H **5 : 3** | I **15 : 24** |
| R **35 : 45** | F **30 : 27 : 6** | P **8 : 18** | S **16 : 28 : 24** |

**Check it.** Read the 6 letters in order. Do they spell a word?

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S1-05 (P6-05)

# Missing Terms

★ Name _____

**Topic: Equivalent ratios: finding the missing term**

Find the missing number in each box.
Use the code table.

1. $9 : 8 = 54 : \square$

2. $1 : 3 : 6 = 5 : \square : 30$

3. $8 : \square = 24 : 27$

4. Mei and Kumar share some coins in the ratio 6 : 5. Kumar gets 45. How many coins does Mei get?

   ☐

5. $3 : 2 = \square : 18$

6. Grace and Lily share some cards in the ratio 4 : 11. Grace gets 16. How many cards does Lily get?

   ☐

**Code table**

| | | | | |
|---|---|---|---|---|
| 9 = A | 15 = E | 16 = J | 18 = K | 27 = U |
| 44 = P | 48 = T | 54 = C | 96 = Q | |

**Check it.** Write the letters in order. Do they spell a word?

☐☐☐☐☐☐

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S1-06 (P6-06)

# Share in a Ratio

★ Name _____

**Topic: Ratio: sharing a quantity in a given ratio**

Find each share.
Write the answer in the box.
Use the code table.

**1** Nadia and Siti share $60 in the ratio 3 : 2. How much more does Nadia get than Siti?

**2** Share 64 sweets in the ratio 3 : 1. How many sweets are in the smaller share?

**3** A rope 80 m long is cut into three pieces in the ratio 5 : 4 : 7. How long is the shortest piece?

**4** The ratio of men to women is 7 : 5. There are 72 in all. How many women are there?

**5** Share 210 stickers in the ratio 3 : 7. How many stickers are in the larger share?

**6** Share 140 in the ratio 2 : 3 : 5. What is the largest share?

**Code table**

| | | | | |
|---|---|---|---|---|
| 12 = B | 14 = Q | 16 = A | 20 = N | 21 = G |
| 30 = D | 36 = X | 70 = T | 147 = I | |

**Check it.** Write the letters in order. Do they spell a word?

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S1-07 (P6-07)

# Three Quantities

★★ Name _____

**Topic: Ratio of three quantities in simplest form**

Write each ratio in simplest form.
Put one number in each box.
Use the code table.

| | | |
|---|---|---|
| 1 | The ratio of Tom's money to Ravi's money is 2 : 7. The ratio of Ravi's money to Daniel's money is 7 : 4. Write Tom : Ravi : Daniel. | ☐ : ☐ : ☐ |
| 2 | A box has 28 pens, 20 pencils and 4 rulers. Write pens : pencils : rulers in simplest form. | ☐ : ☐ : ☐ |
| 3 | The ratio of Tom's money to Ravi's money is 5 : 6. The ratio of Ravi's money to Daniel's money is 3 : 1. Write Tom : Ravi : Daniel in simplest form. | ☐ : ☐ : ☐ |
| 4 | A drink has 500 mℓ of juice, 1 ℓ of water and 1 ℓ of syrup. Write juice : water : syrup in simplest form. | ☐ : ☐ : ☐ |
| 5 | A bag has 12 red, 8 blue and 6 green counters. Write red : blue : green in simplest form. | ☐ : ☐ : ☐ |
| 6 | A cake needs 1 kg of flour, 250 g of sugar and 750 g of butter. Write flour : sugar : butter in simplest form. | ☐ : ☐ : ☐ |

**Code table**

| | | |
|---|---|---|
| 1 : 2 : 2 = O | 2 : 4 : 4 = P | 2 : 7 : 4 = D |
| 3 : 1 : 4 = W | 4 : 1 : 3 = R | 5 : 3 : 1 = Z |
| 5 : 6 : 2 = V | 6 : 4 : 3 = U | 7 : 5 : 1 = E |

**Check it.** Write the letters in order. Do they spell a word? ☐☐☐☐☐☐

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S1-08 (P6-08)

# Fractions and Ratios ★★

Name _____

**Topic: Ratio and fraction: how they are linked**

Read each question.
Circle the letter beside the right answer.
Write the six letters in the boxes.

**1** Kumar spent $\frac{4}{5}$ of the money. Find the ratio of the money spent to the money left.

(Z) 4 : 5 (F) 4 : 1 (K) 1 : 4

**2** Ali has $\frac{2}{9}$ of the stickers. Aisha has the rest. Find the ratio of Ali's stickers to Aisha's stickers.

(U) 2 : 9 (A) 2 : 7 (Y) 7 : 2

**3** The ratio of red beads to blue beads is 1 : 5. What fraction of the beads is red?

(C) $\frac{1}{5}$ (T) $\frac{1}{6}$ (L) $\frac{5}{6}$

**4** The ratio of cats to dogs is 5 : 7. What fraction of the pets are dogs?

(H) $\frac{7}{12}$ (S) $\frac{5}{12}$ (E) $\frac{7}{13}$

**5** The ratio of Hui's age to Hui's father's age is 3 : 8. Hui's age is what fraction of the father's age?

(G) $\frac{5}{8}$ (J) $\frac{3}{11}$ (O) $\frac{3}{8}$

**6** Jun's pencil is $\frac{3}{4}$ as long as Farid's pencil. Find the ratio of Jun's pencil to Farid's pencil.

(M) 3 : 4 (R) 3 : 1 (N) 4 : 3

**Check it.** Write the 6 letters in order. Do they spell a word?

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S1-09 (P6-09)

# Colour the Car

★★ Name _____

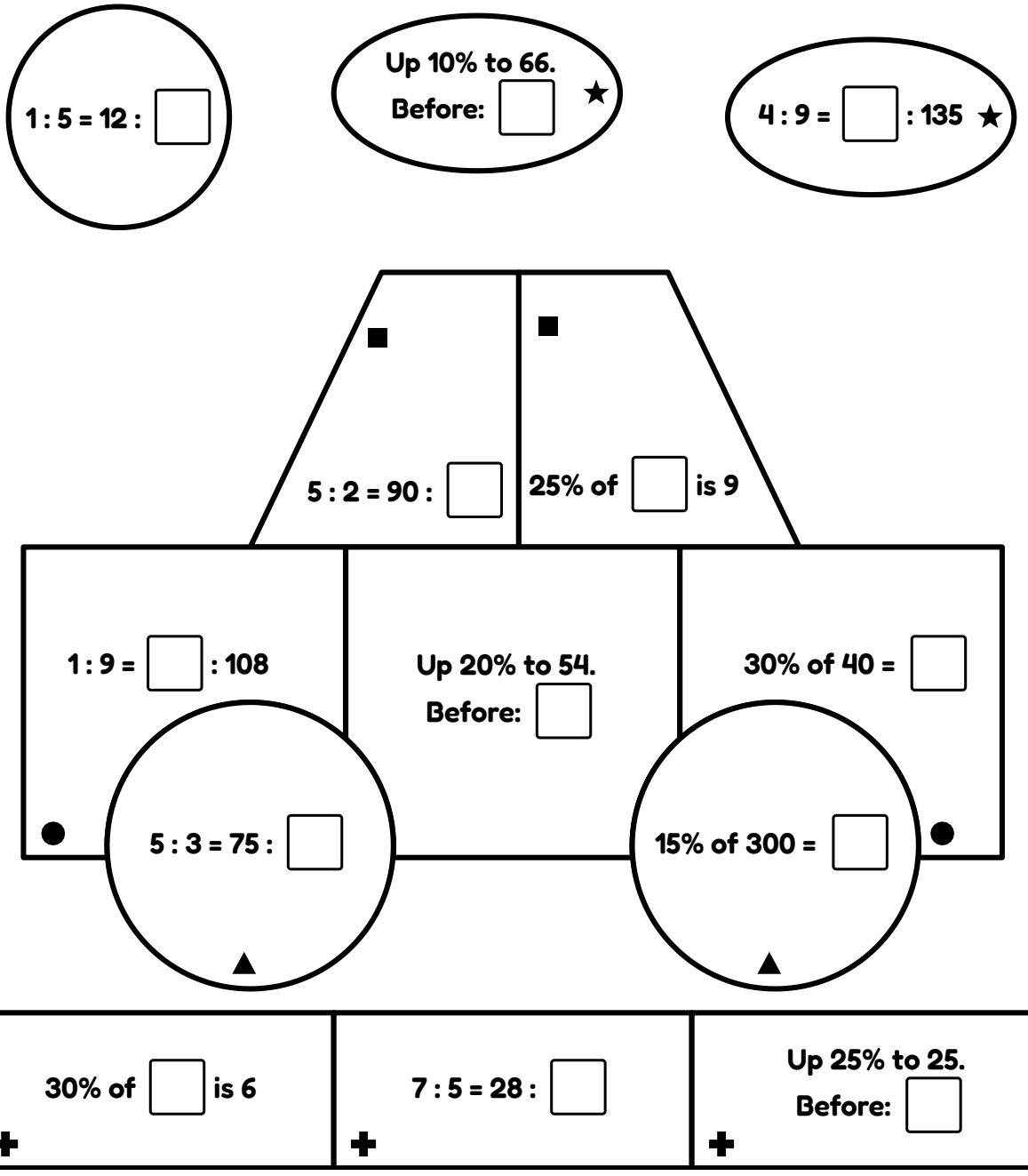
**Topic: Ratio and percentage**

Work out the answer in each part.

Find it in the key.

Colour the part. No crayon? Write Y, R, B, G or O.

| 12 | 20 | 36 | 45 | 60 |
|---|---|---|---|---|
| yellow | red | blue | green | orange |




**Check it.**

Count the parts of each colour. You should have: yellow 2, red 3, blue 2, green 3, orange 3.

Parts with the same small mark have the same colour: ▲ ■ ★ ✚ ●

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S1-10 (P6-10)

# Find the Whole

★ Name _____

**Topic: Percentage: finding the whole from a part**

Find the whole in each question.
Write the answer in the box.
Use the code table.

| | | |
|---|---|---|
| 1 | Ben saved $128, which is 20% of Ben's pocket money. How much pocket money did Ben get? | |
| 2 | 15% of a number is 24. Find the number. | |
| 3 | 50% of a ribbon is 95 cm long. Work out the whole length in cm. | |
| 4 | Boys make up 5% of the people in a hall. 33 boys are there. Find the number of people in the hall. | |
| 5 | A bike is on sale at 35% of its usual price. The sale price is $126. What is the usual price? | |
| 6 | A tank is 35% full. It holds 168 ℓ of water. How much water does the full tank hold in ℓ? | |

**Code table**

| | | | | |
|---|---|---|---|---|
| 136 = P | 160 = R | 190 = U | 312 = J | 360 = T |
| 480 = Y | 627 = Z | 640 = F | 660 = I | |

**Check it.** Write the letters in order. Do they spell a word? ☐☐☐☐☐☐

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S1-11 (P6-11)

# Percent Up and Down

★ Name _____

**Topic: Percentage increase and decrease**

Start at START. Move up, down, left or right.
Step only on boxes that show a 25% increase.
Do not step on a box twice.

| | | | |
|---|---|---|---|
| Up from 144 to 180 | Up from 88 to 110 | Up from 220 to 275 | Up from 116 to 145 |
| Down from 220 to 165 | START<br>Up from 16 to 20 | Up from 90 to 120 | Up from 8 to 10 |
| Down from 80 to 32 | Up from 96 to 120 | Up from 105 to 168 | Up from 120 to 150 |
| Down from 80 to 20 | Up from 165 to 264 | FINISH<br>Up from 236 to 295 | Up from 224 to 280 |

**Check it.** Does your line reach FINISH? Count its boxes: are there 8? Is every box a 25% increase?

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S1-12 (P6-12)

# Percent Stories

★★ Name _____

**Topic: Percentage: finding the whole and the percentage change**

Work out each answer.
Write it in the box.
Use the code table.

| | | |
|---|---|---|
| 1 | A phone cost $800. In a sale the price fell to $760. Find the percentage decrease. | |
| 2 | The price of a book rose by 40%. The new price is $560. What was the old price in dollars? | |
| 3 | A plant was 60 cm tall. It grew to 78 cm. Find the percentage increase. | |
| 4 | A shop sold 60% of its mangoes and had 168 mangoes left. Find the number of mangoes at the start of the day. | |
| 5 | A bag is sold at a 30% discount. The sale price is $224. What was the usual price in dollars? | |
| 6 | Alex has 30% more stickers than Jenny. Alex has 442 stickers. How many stickers does Jenny have? | |

**Code table**

| | | | | |
|---|---|---|---|---|
| 5 = S | 10 = X | 18 = Z | 30 = U | 228 = B |
| 320 = R | 340 = E | 400 = Q | 420 = A | |

**Check it.** Write the letters in order. Do they spell a word? ☐☐☐☐☐☐

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S1-13 (P6-13)

# Letters for Numbers

★ Name _____

**Topic: Algebra: writing expressions with a letter**

A letter stands for a number.
Circle the letter beside the right expression.
Write the six letters in the boxes.

1. Nadia shares $k$ stickers equally among 3 friends. How many stickers does each friend get?

   (S) $k \div 3$ (X) $3k$ (E) $3 \div k$

2. A pen costs \$$n$. A ruler costs \$4 more than the pen. How many dollars does the ruler cost?

   (J) $n - 4$ (N) $4n$ (T) $n + 4$

3. A box holds $y$ pencils. How many pencils are in 6 boxes?

   (F) $y + 6$ (U) $6y$ (W) $y - 6$

4. Wei is $t$ years old. Farid is 7 years younger. How old is Farid?

   (D) $t - 7$ (M) $t + 7$ (C) $t \div 7$

5. Hui has $p$ sweets. Hui eats 9 of them. How many sweets are left?

   (Z) $p + 9$ (G) $9p$ (I) $p - 9$

6. One book costs \$$w$. How many dollars do 8 books cost?

   (Y) $w \div 8$ (K) $w + 8$ (O) $8w$

**Check it.** Write the 6 letters in order. Do they spell a word?

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S1-14 (P6-14)

# Simplify

Name _____

**Topic: Algebra: simplifying expressions with one letter**

Simplify each expression.
Find the card with the simplest form.
Write the card letter in the box. Use each card once.

**Items**

| | |
|---|---|
| (1) $9p + 5p + 12$ ☐ | (2) $8a - 2a + 5$ ☐ |
| (3) $9k + 10 + 5k + 4$ ☐ | (4) $4y + y + 12$ ☐ |
| (5) $7w + 5 - 2w$ ☐ | (6) $9t - 8t + 6t + 12$ ☐ |

**Cards**

| | | | |
|---|---|---|---|
| (S) $7t + 12$ | (J) $12p + 14$ | (W) $14p + 12$ | (A) $6a + 5$ |
| (R) $5y + 12$ | (U) $5w + 5$ | (X) $12y + 5$ | (L) $14k + 14$ |

**Check it.** Read the 6 letters in order. Do they spell a word?

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S1-15 (P6-15)

# Substitute

★★ Name _____

**Topic: Algebra: finding the value of an expression**

Work out the value of each half domino.
The right half of a domino has the same value as the left half of the next domino.
Write the letters in order, from START.

| | | |
|---|---|---|
| A | $3p + 27$ when $p = 7$ | $2p + 4$ when $p = 7$ |
| B | $5k + 18$ when $k = 7$ | START $w \div 4 + 35$ when $w = 20$ |
| O | $a \div 4 + 31$ when $a = 36$ | $k \div 2 + 8$ when $k = 18$ |
| T | $k \div 4 + 13$ when $k = 16$ | $4t + 12$ when $t = 9$ |
| Y | $t \div 3 + 35$ when $t = 12$ | $5k + 23$ when $k = 6$ |
| N | $5p - 12$ when $p = 6$ | $t \div 2 + 34$ when $t = 10$ |

**Write the letters in order, from START.**

| | | | | | |
|---|---|---|---|---|---|
| | | | | | |

**Check it.** Does the loop close, back at START? Read the 6 letters in order: do they spell a word?

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S1-16 (P6-16)

# Find the Letter

★★ Name _____

**Topic: Algebra: solving simple equations**

Find the value of the letter in each equation.
Write the number in the box.
Use the code table.

**1** $9w - 14 = 40$

$w =$ ☐

**2** Think of a number $p$. Multiply it by 2 and then add 15. The answer is 67. Find $p$.

$p =$ ☐

**3** Hana buys 6 pens at $\$k$ each and a ruler for $18. The total is $30. Find $k$.

$k =$ ☐

**4** $5n = 145$

$n =$ ☐

**5** $5y - 2y = 9$

$y =$ ☐

**6** $2t + 12 = 58$

$t =$ ☐

**Code table**

| | | | | |
|---|---|---|---|---|
| 2 = A | 3 = T | 5 = Z | 6 = C | 8 = P |
| 23 = Y | 26 = R | 29 = F | 41 = L | |

**Check it.** Write the letters in order. Do they spell a word? ☐☐☐☐☐☐

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S1-17 (P6-17)

# Round and Round

★ Name _____

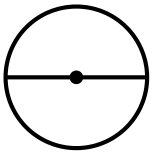
**Topic: Circles: circumference**

Take π as 22/7.

Find the distance round each circle.

Write the answer in the box. Use the code table.

**1**



**d = 56 cm**

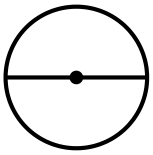
Find the circumference.

_____ **cm**

**2** A round clock has a radius of 14 cm. How long is its edge, in cm?

_____ **cm**

**3**



**d = 49 cm**

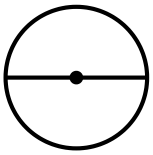
Find the circumference.

_____ **cm**

**4** A circular pond has a diameter of 7 m. How far is it round the pond, in m?

_____ **m**

**5**



**d = 21 cm**

Find the circumference.

_____ **cm**

**6** A wheel has a diameter of 14 cm. How far does it roll in one turn, in cm?

_____ **cm**

**Code table**

| | | | | |
|---|---|---|---|---|
| 7 = F | 22 = S | 28 = L | 29 = Z | 44 = N |
| 66 = I | 88 = O | 154 = U | 176 = C | |

**Check it.** Write the letters in order. Do they spell a word?

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S1-18 (P6-18)

# Circle Areas

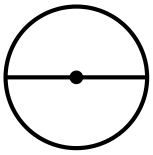
★★ Name _____

**Topic: Circles: area**

Take π as 3.14.

Find the area of each circle.

Write the answer in the box. Use the code table.

**1** 

**d = 8 cm**

Find the area.

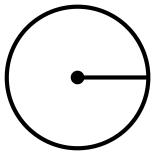
_____ **cm²**

**2** The circumference of a circle is 43.96 cm. Find its area.

_____ **cm²**

**3** A round pond has a radius of 8 m. Find the area of the pond.

_____ **m²**

**4** 

**r = 10 cm**

Find the area.

_____ **cm²**

**5** A round pond has a radius of 2 m. Find the area of the pond.

_____ **m²**

**6** A round pizza has a diameter of 6 cm. Find its area.

_____ **cm²**

**Code table**

| | | | | |
|---|---|---|---|---|
| 12.56 = E | 28.26 = N | 31.4 = J | 50.24 = F | 100.48 = M |
| 153.86 = R | 200.96 = O | 314 = Z | 803.84 = Y | |

**Check it.** Write the letters in order. Do they spell a word? _____

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S1-19 (P6-19)

# Half and Quarter

★★ Name _____

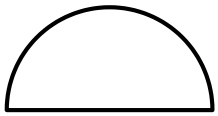
**Topic: Semicircles and quarter circles: area and perimeter**

Take π as 22/7.

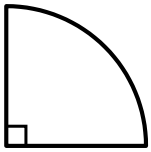
Find the area or the perimeter of each shape, as the word says.

Write the card letter in the box. Use each card once.

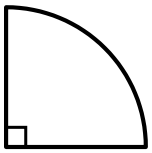
**Items**

1  **Area** ☐

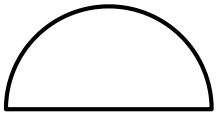
d = 7 cm

2  **Area** ☐

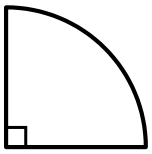
r = 7 cm

3  **Perimeter** ☐

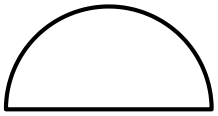
r = 7 cm

4  **Perimeter** ☐

d = 28 cm

5  **Perimeter** ☐

r = 28 cm

6  **Area** ☐

d = 28 cm

**Cards**

| | | | |
|---|---|---|---|
| H **19.25 cm²** | S **44 cm** | Q **25 cm²** | G **72 cm** |
| Y **308 cm²** | R **100 cm** | U **38.5 cm²** | N **25 cm** |

**Check it.** Read the 6 letters in order. Do they spell a word?

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S1-20 (P6-20)

# Curves and Corners

★★ Name _____

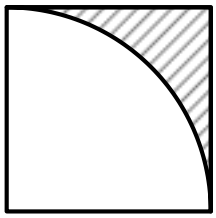
**Topic: Composite figures with semicircles and quarter circles**

Take π as 22/7.

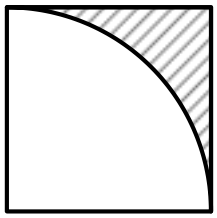
Find the area of each figure. For a hatched figure, find the hatched part.

Put the areas in order, smallest first. Write the letters.

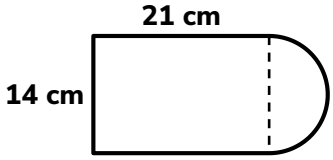
**L**



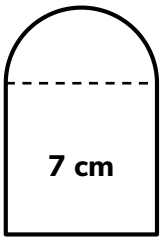

**P**



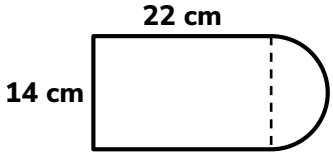

**I**



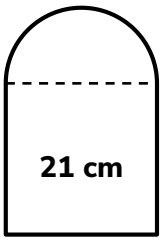

**O**




**C**




**E**




**Write the letters in order, smallest area first.**

| | | | | | |
|---|---|---|---|---|---|
| | | | | | |

**Check it.** Read the 6 letters in order. Do they spell a word?

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